

MASBATE, Philippines

WMO: 985430

Lat: 12.3663N

Lon: 123.6292E

Elev: 8

StdP: 101.23

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.2	23.8	20.8	15.5	25.4	21.5	16.1	25.6	8.9	26.8	8.0	27.3	1.8	0	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	6.5	33.8	28.0	33.2	27.7	32.8	27.6	28.7	32.5	28.4	32.1	28.1	31.8	2.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.7	23.7	31.6	27.3	23.1	31.2	27.1	22.9	31.0	93.3	32.7	91.8	32.4	90.5	32.0	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.5	5.5	22.0	35.3	1.1	1.2	21.2	36.2	20.6	36.8	20.0	37.5	19.2	38.4		
(5)			20.7	29.6	1.4	0.7	19.7	30.1	18.8	30.5	18.1	30.8	17.1	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	27.0	27.3	28.0	29.1	29.5	29.2	28.6	28.7	28.6	28.6	28.3	27.6		
	DBStd	1.24	1.06	1.00	1.00	0.90	0.95	1.10	1.04	0.99	0.95	0.92	0.95	1.07		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6708	526	486	558	572	605	575	576	581	558	576	550	545		
	CDD18.3	3667	268	252	300	322	347	325	318	323	308	318	300	287		
	CDH23.3	40493	2353	2373	3140	3826	4336	3940	3632	3744	3508	3547	3274	2820		
	CDH26.7	14456	539	655	1028	1565	1924	1645	1322	1417	1272	1257	1083	747		
(14) Wind	WSAvg	2.2	2.4	2.4	2.2	2.1	2.1	2.0	2.2	2.3	2.0	1.8	2.0	2.4		
(15) Precipitation	PrecAvg	1793	165	65	57	35	125	146	171	195	206	213	223	192		
	PrecMax	2386	543	149	135	98	342	288	344	435	351	367	366	434		
	PrecMin	1150	16	3	1	5	3	9	5	73	55	74	74	25		
	PrecStd	351	151	41	39	31	113	67	89	107	80	105	93	118		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	32.4	33.1	34.1	34.3	34.3	33.6	33.6	33.4	33.5	33.1	32.2		
		MCWB	27.2	27.1	27.5	28.0	28.1	28.2	28.0	27.6	27.7	28.1	28.0	27.7		
	2%	DB	30.9	31.5	32.1	33.2	33.7	33.4	32.6	32.8	32.7	32.4	32.2	31.5		
		MCWB	26.7	26.8	27.1	27.6	27.9	28.0	27.5	27.4	27.5	27.6	27.7	27.3		
	5%	DB	30.2	30.8	31.5	32.6	33.0	32.8	32.0	32.1	32.0	31.9	31.6	30.9		
		MCWB	26.4	26.5	26.8	27.3	27.7	27.8	27.2	27.2	27.2	27.3	27.5	27.0		
	10%	DB	29.5	30.1	30.9	32.0	32.4	32.0	31.2	31.5	31.2	31.2	31.0	30.1		
		MCWB	26.0	26.1	26.5	27.0	27.5	27.5	27.0	27.1	27.1	27.1	27.2	26.7		
	0.4%	WB	28.0	28.1	28.4	28.7	29.1	29.1	28.7	28.5	28.6	28.7	28.7	28.4		
		MCDB	30.9	31.3	32.0	32.8	33.1	33.1	32.4	32.2	32.2	32.3	31.9	31.2		
	2%	WB	27.2	27.3	27.7	28.2	28.6	28.6	28.1	28.0	28.1	28.1	28.2	27.8		
		MCDB	30.1	30.6	31.3	32.3	32.5	32.4	31.5	31.7	31.6	31.6	31.4	30.6		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	26.7	26.9	27.2	27.8	28.2	28.2	27.7	27.6	27.7	27.7	27.7	27.3		
		MCDB	29.5	30.1	30.7	31.8	32.1	31.9	31.0	31.2	31.0	31.0	30.9	30.1		
	10%	WB	26.2	26.3	26.8	27.4	27.8	27.7	27.2	27.2	27.2	27.3	27.3	26.9		
		MCDB	28.8	29.4	30.2	31.2	31.6	31.3	30.4	30.6	30.5	30.5	30.3	29.5		
	5% DB	MDBR	5.3	6.1	6.4	6.8	6.5	6.1	5.8	5.9	5.9	6.1	5.8	5.4		
		MCDBR	6.6	7.2	7.2	7.5	7.2	7.0	6.7	6.8	6.9	6.9	6.7	6.4		
(36) Mean Daily Temperature Range	5% WB	MCWBR	3.6	3.7	3.6	3.5	3.2	3.2	3.0	3.0	3.3	3.3	3.5	3.4		
		MCDBR	6.1	6.7	6.9	7.2	6.8	6.7	6.5	6.5	6.6	6.6	6.4	6.0		
	5% WB	MCWBR	3.6	3.8	3.7	3.6	3.4	3.4	3.4	3.3	3.5	3.5	3.6	3.5		
		MCDBR	6.1	6.7	6.9	7.2	6.8	6.7	6.5	6.5	6.6	6.6	6.4	6.0		
(40) Clear-Sky Solar Irradiance	taub		0.447	0.432	0.434	0.433	0.431	0.433	0.436	0.455	0.450	0.455	0.442	0.454		
	taud		2.368	2.406	2.414	2.418	2.448	2.458	2.450	2.380	2.388	2.368	2.403	2.347		
	Ebn at Noon		855	884	887	878	865	854	853	848	859	851	853	838		
	Edh at Noon		121	121	122	121	115	112	114	124	123	124	116	122		
(44) All-Sky Solar Radiation	RadAvg		4.10	5.01	5.71	6.39	5.93	5.37	4.91	5.06	4.99	4.72	4.47	3.92		
	RadStd		0.46	0.56	0.60	0.38	0.51	0.33	0.36	0.28	0.46	0.30	0.39	0.34		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+0.26	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page